Intake of fatty acids in general populations worldwide does not meet dietary recommendations to prevent coronary heart disease: a systematic review...

AIM: To systematically review data from different countries on population intakes of total fat, saturated fatty acids (SFA) and polyunsaturated fatty acids (PUFA), and to compare these to recommendations from the Food and Agriculture Organization of the United Nations/the World Health Organization (FAO/WHO). METHODS: Data from national dietary surveys or population studies published from 1995 were searched via MEDLINE, Web of Science and websites of national public health institutes. RESULTS: Fatty acid intake data from 40 countries were included. Total fat intake ranged from 11.1 to 46.2 percent of energy intake (% E), SFA from 2.9 to 20.9% E and PUFA from 2.8 to 11.3% E. The mean intakes met the recommendation for total fat (20-35% E), SFA (<10% E) and PUFA (6-11% E) in 25, 11 and 20 countries, respectively. SFA intake correlated with total fat intake ($r = 0.76$, $p < 0.01$) but not with PUFA intake ($r = 0.03$, $p = 0.84$). Twenty-seven countries provided data on the distribution of fatty acids intake. In 18 of 27 countries, more than 50% of the population had SFA intakes >10% E and in 13 of 27 countries, the majority of the population had PUFA intakes <6% E. CONCLUSIONS: In many countries, the fatty acids intake of adults does not meet the levels that are recommended to prevent chronic diseases. The relation between SFA and PUFA intakes shows that lower intakes of SFA in the populations are not accompanied by higher intakes of PUFA, as is recommended for preventing coronary heart disease.